

**BOARD QUESTION PAPER:
MARCH 2015 Physics**

Time: 3 Hours

Total Marks: 70

Note:

- i. All questions are compulsory.
- ii. Answers to the two sections are to be written in the same answer book.
- iii. Figures to the right hand side indicate full marks.
- iv. Write balanced chemical equations and draw neat and labelled diagrams, wherever necessary.
- v. Use of logarithmic table is allowed.
- vi. Answer to every question must be started on a new page.

SECTION – I

Q.1. Select and write the most appropriate answer from the given alternatives for each sub-question: [7]

(i) The period of a conical pendulum in terms of its length (l), semi vertical angle (θ) acceleration due to gravity (g) is _____.

- (a) $\frac{1}{2\pi} \sqrt{\frac{l \cos \theta}{g}}$ (b) $\frac{1}{2\pi} \sqrt{\frac{l \sin \theta}{g}}$ (c) $4\pi \sqrt{\frac{l \cos \theta}{4g}}$ (d) $4\pi \sqrt{\frac{l \tan \theta}{g}}$

(ii) The kinetic energy of a rotating body depends upon _____.

- (a) Distribution of mass only
(b) Angular speed only
(c) Distribution of mass and angular speed
(d) Angular acceleration only

(iii) If the metal bob of a simple pendulum is replaced by a wooden bob of the same size, then its time period will _____.

- (a) Increase (b) Remain same (c) decreases (d) first increase and then decreases

(iv) The graph between applied force and change in the length of wire within elastic limit is _____.

- (a) Straight line with positive slope.
(b) Straight line with negative slope.
(c) Curve with positive slope.
(d) Curve with negative slope.

(v) When a longitudinal wave is incident at the boundary of denser medium, then, _____.

- (a) Compression reflects as a compression.
- (b) Compression reflects as a rarefaction.
- (c) Rarefaction reflects as a compression.
- (d) Longitudinal wave reflects as transverse wave.

(vi) The dimensions of universal gravitational constant are _____.

- (a) $[L^1M^0T^0]$
- (b) $[L^2M^1T^0]$
- (c) $[L^{-1}M^1T^{-2}]$
- (d) $[L^3M^{-1}T^{-2}]$

(vii) Two copper spheres of radii 6cm and 12 cm respectively are suspended in an evacuated enclosure. Each of them are at a temperature 15°C above the surroundings. The ratio of their rate of loss of heat is _____.

- (a) 2:1
- (b) 1:4
- (c) 1:8
- (d) 8:1

Solution 1:

(i) (c) $4\pi\sqrt{\frac{l\cos\theta}{4g}}$

(ii) (c) Distribution of mass and angular speed.

(iii) (b) Remains same.

(iv) (a) Straight line with positive slope.

(v) (a) compression reflects as a compression

(vi) (d) $[L^3M^{-1}T^{-2}]$

(vii) (b) 1:4

Q2. Answer any SIX of the following:

[12]

(i) In a circular motion, assuming $\vec{v} = \vec{\omega} \times \vec{r}$, obtain an expression for the resultant acceleration of a particle in terms of tangential and radial component.

(ii) Explain why an astronaut in an orbiting satellite has a feeling of weightlessness.

(iii) State theorem of parallel axes and theorem of perpendicular axes about moment of inertia.

(iv) State: (a) Wien's displacement law, and (b) first law of thermodynamics.

(v) A particle in S.H.M. has a period of 2 seconds and amplitude of 10cm. Calculate the acceleration when it is at 4 cm from its positive extreme position.

(vi) The surface tension of water at 0°C is 75.5 dyne/cm. Calculate surface tension of water at 25°C .
(α for water = $2.7 \times 10^{-3}/^\circ\text{C}$)

(vii) The spin dryer of a washing machine rotating at 15 r.p.s slows down to 5 r.p.s. after making 50 revolutions. Find its angular acceleration.

(viii) Calculate the period of revolution of Jupiter around the Sun. The ratio of the radius of Jupiter's orbit to that of the Earth's orbit is 5. (Period of revolution of the Earth is 1 year)

Solution 2:

Let the particle be moving around a circular path of constant radius r . If it speeds up or slows down, its motion is non-uniform and both the angular speed (ω) and the linear speed (v) change with time. Thus, at any instant, ω , v and r are related as:

$$\vec{v} = \vec{\omega} \times \vec{r}$$

On differentiating, we get,

$$\frac{d\vec{v}}{dt} = \frac{d(\vec{\omega} \times \vec{r})}{dt}$$

$$\Rightarrow \frac{d\vec{v}}{dt} = \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt}$$

$$\text{But } \frac{d\vec{\omega}}{dt} = \vec{\alpha} \text{ and } \frac{d\vec{r}}{dt} = \vec{v}$$

$$\therefore \frac{d\vec{v}}{dt} = \vec{\alpha} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt}$$

$$\vec{a} = \vec{a}_1 + \vec{a}_2$$

Where, $\vec{a}_1$ = tangential component

$\vec{a}_2$ = Radial component

(ii) An astronaut of mass m in a satellite is moving with constant speed along the orbit. When an astronaut is in an orbiting satellite, the satellite as well as the astronaut are attracted to the centre of the earth and both fall towards earth with same acceleration. This acceleration is same as g at the satellite. Thus, the astronaut is unable to exert weight on the floor of the satellite, in turn the satellite does not provide normal reaction on the astronaut. Therefore, the astronaut feels weightlessness.

(iii) The theorem of parallel axis states that the moment of inertia (I) of a body about any axis is equal to the sum of its moment of inertia (I_{CM}) about a parallel axis through the centre of mass and the product of mass M of the body and the square of the perpendicular distance between the two parallel axes.

The theorem of perpendicular axes about moment of inertia states that the moment of inertia of a plane lamina about an axis perpendicular to its plane is equal to the sum of its moment of inertia about two mutually perpendicular axes in its plane and through the point of intersection of the perpendicular axis and the lamina.

(iv) (a) Wien's displacement law states that the wavelength for which emissive power of a black body is maximum is inversely proportional to the absolute temperature of the black body

$$\lambda_{\max} \propto \frac{1}{T}$$

$$\therefore \lambda_{\max} = b \times \frac{1}{T}$$

Where, b = Wien's constant.

(b) The first law of thermodynamics between any two states of equilibrium states that the change in internal energy is equal to the difference of the heat transfer into the system and work done by the system. Thus, in simpler terms, energy can be changed from one form to another but it cannot be created or destroyed. Hence, energy is always conserved.

(v) Given: $A = 10 \text{ cm}$, $T = 2 \text{ sec}$

$$a = (2\pi f)^2 \cdot x$$

$$= \left(2 \times \frac{22}{7} \times f \right)^2 \cdot x$$

$$f = \frac{1}{T} - \frac{1}{2}$$

$$= \left(2 \times \frac{22}{7} \times \frac{1}{2} \right)^2 \cdot x$$

$$x = A - 4$$

$$= 10 - 4$$

$$= 6 \text{ cm}$$

$$a = \left(2 \times \frac{22}{7} \times \frac{1}{2} \right)^2 \cdot 6$$

$$\Rightarrow a = \frac{22 \times 22}{49} \times 6 = \frac{2904}{49}$$

$$\Rightarrow a = 59.2 \text{ m/s}^2$$

(vi) Given:

$$T_0 = 75.5 \text{ dyne/cm}, \alpha = 0.0027 / ^\circ \text{C}, \theta = 25^\circ \text{C}$$

Then,

$$T = T_0 (1 - \alpha \theta)$$

$$= 75.5 [1 - (0.0027)(25)]$$

$$= 75.5 [1 - 0.0675]$$

$$= 75.5 (0.9325)$$

$$= 70.40 \text{ dyne/cm}$$

Thus, the surface tension of water at 25°C is 70.40 dyne/cm .

(vii) Initial angular velocity, $\omega_0 = 2\pi \times 15 \text{ rad/s}$

Final angular velocity, $\omega = 2\pi \times 5 \text{ rad/s}$

Angular displacement, $\theta = 2\pi \times 50 \text{ rad}$.

We know that,

$$\omega^0 = \omega_0^2 + 2\alpha\theta$$

$$\Rightarrow (2\pi \times 5)^2 = (2\pi \times 15)^2 + 2\alpha \times (2\pi \times 50)$$

$$\Rightarrow 100\pi^2 = 900\pi^2 + 200\pi\alpha$$

$$\Rightarrow 200\pi\alpha = -800\pi^2$$

$$\Rightarrow \alpha = -12.56 \text{ rad/s}^2$$

The negative sign indicates that the angular acceleration is directed opposite to the angular velocity.

(viii) Given: $r_J:r_E = 5$, $T_E = 1 \text{ year}$

We know that,

$$T\alpha r^{3/2}$$

$$\therefore \frac{T_J}{T_E} = \left(\frac{r_J}{r_E} \right)^{3/2}$$

$$\Rightarrow T_J = (5)^{3/2} \times 1$$

$$= 11.18 \text{ year}$$

Q.3. Answer any THREE of the following:

[9]

(i) Derive an expression for excess pressure inside a drop of liquid.

(ii) Explain what is Doppler Effect in sound and state its any 'four' applications.

(iii) Calculate the average molecular kinetic energy:

(a) Per kilomole, (b) per kilogram, of oxygen at 27°C .

($R = 8320 \text{ J/K mole K}$, Avogadro's number = $6.03 \times 10^{26} \text{ molecules/k mole}$)

(iv) A uniform steel rod of 5mm^2 cross section is heated from 0°C to 25°C . Calculate the force which must be exerted to prevent it from expanding. Also, calculate strain.

(α for steel = $12 \times 10^{-6}/^\circ\text{C}$ and Y for steel = $20 \times 10^{10} \text{ N/M}^2$)

Solution 3:

(i) Let us consider a liquid drop and let the outside the pressure P_0 and inside pressure be p_i , such that the excess pressure is $P_i - P_0$.

Now,

Let the radius of the drop increases from r to $r + \Delta r$, where Δr is very small, so that pressure inside the drop remains almost constant.

$$\text{Initial surface area } (A_1) = 4\pi r^2$$

$$\text{Final surface area } (A_2) = 4\pi (r + \Delta r)^2$$

$$= 4\pi(r^2 + 2r\Delta r + \Delta r^2)$$

$$= 4\pi r^2 + 8\pi r\Delta r + 4\pi \Delta r^2$$

As Δr is very small, Δr^2 is neglected $4\pi \Delta r^2 \approx 0$

$$\text{Increase in surface area } (dA) = A_2 - A_1 = 4\pi r^2 + 8\pi r\Delta r - 4\pi r^2$$

So, increase in surface area $(dA) = 8\pi r\Delta r$

Work done to increase the surface area $(dA) = 8\pi r\Delta r$ is extra energy.

So,

$$dW = TdA$$

$$dW = T \times 8\pi r\Delta r \dots\dots(1)$$

This is equal to the product of force and distance Δr .

$$dF = (P_i - P_0)4\pi r^2$$

So, the increase in radius of the bubble is Δr .

$$dW = dF\Delta r = (P_i - P_0)4\pi r^2 \times \Delta r \dots\dots(ii)$$

From eq(i) and(ii), we get

$$(P_i - P_0)4\pi r^2 \times \Delta r = T \times 8\pi r\Delta r$$

$$(P_i - P_0) = \frac{2T}{r}$$

Hence, this is the Laplace theorem.

(ii) When there is a relative motion between a source of sound and an observer, there is apparent change in frequency of a sound emitted by a source and as heard by the observer. The Doppler Effect causes the received frequency of a source (how it is perceived when it gets to its destination) to differ from the sent frequency if there is motion i.e., increasing or decreasing the distance between the source and the receiver. This effect is readily observable by the variation in the pitch of sound between a moving source and a stationary observer.

Four applications of Doppler Effect include:

- The handheld radar guns used by police to check for speeding vehicles rely on the Doppler Effect.
- Meteorologists use a similar principle to read weather events. In this case, the stationary transmitter is located in a weather station and the moving object being studied is a storm system.
- It is used to determine the speed of rotation of the sun.
- A traditional echocardiogram uses sound waves to produce images of the heart.

(iii) (a) Average molecular K.E. per k mole at 27°C ,

$$\text{K.E.} = \frac{3}{2} RT$$

$$= \frac{3}{2} \times 8320 \times 300$$

$$= 3.744 \times 10^6 \text{ J}$$

(b) Average molecular K.E. per kilogram

$$= \frac{\text{K.E. per mole}}{M}$$

$$= \frac{3.744 \times 10^6}{32}$$

$$= 1.17 \times 10^5 \text{ J}$$

(iv)

$$F = Y_{\text{steel}} \times \alpha_{\text{steel}} \times \Delta T \times A$$

$$= 20 \times 10^{10} \times 12 \times 10^{-6} \times 25 \times 5 \times 10^{-6}$$

$$= 30000 \times 10^{10} \times 10^{-12}$$

$$= 300 \text{ N}$$

$$\text{Strain} = \alpha_{\text{steel}} \times \Delta T$$

$$= 12 \times 10^{-6} \times 25$$

$$= 300 \times 10^{-6}$$

$$= 3 \times 10^{-4}$$

Question 4.

[7]

What are forced vibration and resonance? Show that only odd harmonics are present in an air column vibrating in a pipe closed at one end.

A stretched wire emits a fundamental note of frequency 256 Hz. Keeping the stretching force constant and reducing the length of wire by 10 cm, the frequency becomes 320 Hz. Calculate the original length of wire.

OR

Obtain an expression for potential energy of a particle performing simple harmonic motion. Hence, evaluate the potential energy (a) at mean position and (b) at extreme position.

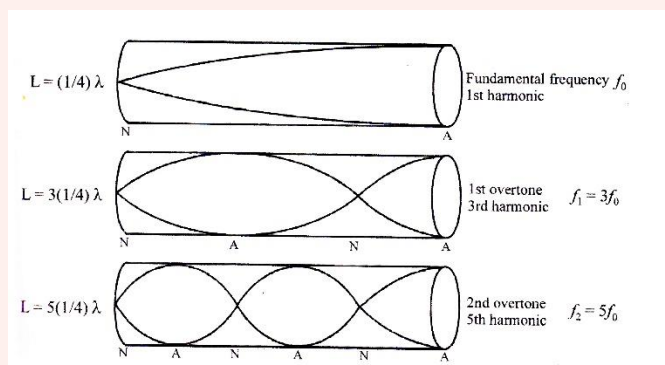
A horizontal disc is freely rotating about a transverse axis passing through its centre at the rate of 100 revolutions per minute. A 20 gram blob of wax falls on the disc and sticks to the disc at a distance of 5 cm from its axis. Moment of inertia of the disc about its axis passing through its centre of mass. Calculate the new frequency of rotation of the disc.

Solution 4:

Forced Vibrations: The vibrations of body under action of external; periodic force in which body vibrates with frequency equal to frequency of external periodic force other than its natural frequency are called forced vibrations.

Resonance: If an object is being forced to vibrate at its natural frequency, resonance will occur and large amplitude vibrations can be observed. The resonant frequency is f_0 .

Modes of vibration of air column in a pipe closed at one end:



Thus, it can be seen that only odd harmonics are present as overtone in a mode of vibration of air column closed at one end.

Numerical: The fundamental frequency of vibration (the simplest mode with one vibrating segment) is given by the relation:

$$f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

Where, f is the frequency in Hz. T is the tension in the string in newton, μ is the mass per unit length of the string (linear density) in kg/m, and L is the length of the string in meters.

Now, the linear density remains the same as there is no chnge in the material of the wire, also, the tension remains constant.

Hence,

$$256 = \frac{1}{2L} \cdot \sqrt{\frac{T}{\mu}}$$

$$\Rightarrow L = \frac{1}{2(256)} \cdot \sqrt{\frac{T}{\mu}} = \frac{1}{512} \cdot \sqrt{\frac{T}{\mu}}$$

Again,

$$\begin{aligned}
 320 &= \frac{1}{2(L-0.1)} \cdot \sqrt{\frac{T}{\mu}} \\
 \Rightarrow L-0.1 &= \frac{1}{2(320)} \cdot \sqrt{\frac{T}{\mu}} \\
 \Rightarrow \frac{1}{512} \cdot \sqrt{\frac{T}{\mu}} - 0.1 &= \frac{1}{640} \cdot \sqrt{\frac{T}{\mu}} \\
 \Rightarrow \sqrt{\frac{T}{\mu}} \left[\frac{1}{512} - \frac{1}{640} \right] &= 0.1 \\
 \Rightarrow \sqrt{\frac{T}{\mu}} [0.00039] &= 0.1 \\
 \Rightarrow \sqrt{\frac{T}{\mu}} &= 256
 \end{aligned}$$

Thus,

$$\begin{aligned}
 L &= \frac{1}{512} (256) \text{m} \\
 &= 0.5 \text{m} \\
 &= 50 \text{cm}
 \end{aligned}$$

OR

An oscillation particle possesses both types of energies potential as well as kinetic. It possesses potential energy on account of its displacement from equilibrium position.

Potential energy: Consider a particle of mass m executing S.H.M. Let x be its displacement from the equilibrium position at any instant t . Since in S.H.M. the force F acting upon the particle is proportional to and opposite to the displacement x , we have,

$$F = -kx$$

Where, the constant k gives the force per unit displacement. The force can also be expressed in terms of potential energy U of the particle as:

$$F = -\frac{dU}{dx}$$

$$\text{Thus, } \frac{dU}{dx} = kx$$

On integrating, we get,

$$U = \frac{1}{2} kx^2 + C$$

Where C is a constant. If we assume the potential energy zero in the equilibrium position, ie, If $U = 0$ at $x = 0$, then $C = 0$.

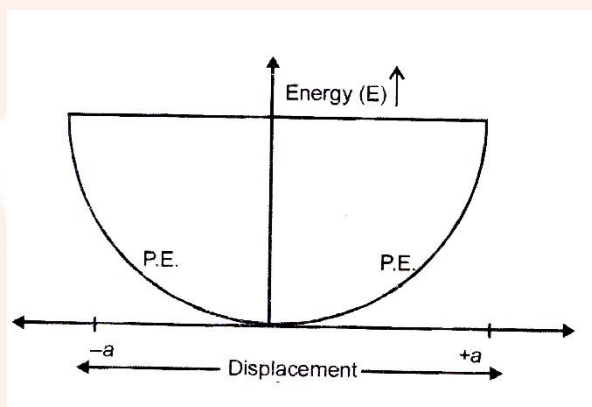
Therefore, $U = \frac{1}{2} kx^2$

Potential energy, $U = \frac{1}{2} m\omega^2 x^2$

Hence, from here it is clear that the potential energy of a particle performing S.H.M. is directly proportional to the square of the displacement ($P.E. \propto x^2$).

(a) Potential energy at mean position: At mean position, velocity of a particle executing S.H.M. is maximum and displacement is minimum i.e., $x = 0$.

(b) Potential energy at extreme position: At extreme position, velocity of a particle executing S.H.M. is minimum and displacement is maximum, i.e., $x = \pm a$.



$$U = \frac{1}{2} ka^2 = \frac{1}{2} m\omega^2 x^2$$

Numerical: Given:

$$n_1 = 100 \text{ r.p.m}$$

$$= \frac{100}{60} = 1.66 \text{ r.p.s}$$

$$m = 20 \text{ gm} = 2 \times 10^{-2} \text{ kg}$$

$$h = r = 5 \text{ cm} = 0.05 \text{ m}$$

$$I_1 = \text{Moment of inertia of the disc about its axis} = 2 \times 10^{-4} \text{ kgm}^2$$

I_1 = Moment of inertia of the disc about same axis with the wax.

To find, $n_2 = ?$

From formula we have,

$$I_2 = I_1 + mr^2$$

$$I_1\omega_1 = I_2\omega_2$$

$$I_1(2\pi n_1) = (I_1 + mr^2)(2\pi n_2)$$

$$I_1 n_1 = (I_1 + mr^2) n_2$$

$$n_2 = \frac{I_1 n_1}{I_1 + mr^2}$$

$$n_2 = \frac{2 \times 10^{-4} \times 1.66}{2 \times 10^{-4} + (2 \times 10^{-2} \times 0.05 \times 0.05)}$$

$$= \frac{2 \times 10^{-4} \times 1.66}{2 \times 10^{-4} + 0.5 \times 10^{-4}}$$

$$= \frac{2 \times 10^{-4} \times 1.66}{2.5 \times 10^{-4}}$$

$$= 1.328$$

$$\cong 1 \text{ r.p.s.}$$

$$\Rightarrow n_2 = 1 \text{ r.p.s.}$$

SECTION – II

Question 5: Select and write the most appropriate answer from the given alternatives for each sub-question: [7]

(i) Electric field intensity in free space at a distance 'r' outside the charged conducting sphere of radius 'R' in terms of surface charge density 'σ' is

(a) $\frac{\sigma}{\epsilon_0} \left[\frac{R}{r} \right]^2$ (b) $\frac{\epsilon_0}{\sigma} \left[\frac{R}{r} \right]^2$ (c) $\frac{R}{r} \left[\frac{\sigma}{\epsilon_0} \right]^2$ (d) $\frac{R}{\sigma} \left[\frac{r}{\epsilon_0} \right]^2$

(ii) Instrument which can measure terminal potential difference as well as electromotive force (e.m.f) is:

(a) Wheatstone's meter bridge (b) Voltmeter (c) Potentiometer (d) Galvanometer

(iii) If the frequency of incident light falling on a photosensitive material is doubled, then the kinetic energy of the emitted photoelectron will be:

(a) Same as its initial value (b) two times its initial value (c) more than two times its initial value (d) less than two times its initial value.

(iv) Linear momentum of an electron in Bohr orbit of H-atom (principal quantum number n) is proportional to:

(a) $\frac{1}{n^2}$ (b) $\frac{1}{n}$ (c) n (d) n^2

(v) In a semiconductor, acceptor impurity is:

(a) Antimony (b) Indium (c) Phosphorous (d) Arsenic

(vi) The power radiated by the linear antenna of length l is proportional to (λ = wavelength):

- (a) $\frac{\lambda}{1}$ (b) $\left(\frac{\lambda}{1}\right)^2$ (c) $\frac{1}{\lambda}$ (d) $\left(\frac{1}{\lambda}\right)^2$

(vii) The numerical aperture of objective of a microscope is 0.12. The limit of resolution, when light of wavelength 600 Å is used to view an object is:

- (a) 0.25×10^{-7} m (b) 2.5×10^{-7} m (c) 25×10^{-7} m (d) 250×10^{-7} m

Solution 5:

(i) (a) $\frac{\sigma}{\epsilon_0} \left[\frac{R}{r} \right]^2$

(ii) (c) Potentiometer

(iii) (c) more than two times its initial value

(iv) (b) $\frac{1}{n}$

(v) (b) Indium

(vi) (d) $\left(\frac{1}{\lambda}\right)^2$

(vii) (c) 25×10^{-7} m

Q.6. Answer any SIX of the following:

[12]

(i) What is a Polaroid? State its two uses.

(ii) Draw a neat and labelled diagram of suspended coil type moving coil galvanometer.

(iii) Define: (a) magnetization (b) magnetic intensity.

(iv) Draw a block diagram of generalized communication system.

(v) A solenoid 3.142 m long and 5.0 cm in diameter has two layers of windings of 500 turns each and carries a current of 5A. Calculate the magnetic moment associated with the coil.

(vi) A circular coil of 300 turns and average area $5 \times 10^{-3} \text{ m}^2$ carries a current of 15A.

Calculate the magnitude of magnetic moment associated with the coil.

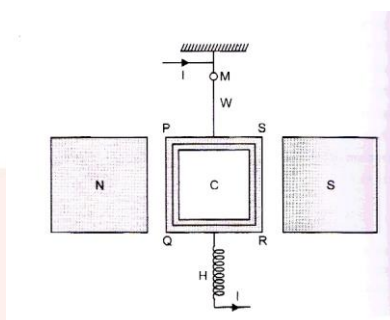
(vii) The magnetic flux through a loop wires according to the relation $\phi = 8t^2 + 6t + C$, where 'C' is constant, ' ϕ ' is in milliweber and 't' is in seconds. What is magnitude of induced e.m.f. in the loop at $t=2$ seconds?

(viii) An electron is orbiting in 5th Bohr orbit. Calculate ionisation energy for this atom, if the ground state energy is -13.6 eV.

Solution 6:

(i) A Polaroid is a synthetic dichroic sheet that can polarize a transmitted beam of normal light because it is composed of long parallel molecules. It only transmits plane – polarized light if these molecules are parallel to the plane of polarization. It is often is sunglasses to eliminate glare and in 3-D motion pictures i.e., holography.

(ii)



PQRS: Rectangular coil

W: Phosphor bronze wire suspension.

M: Plane mirror

H: Phosphor bronze helical spring

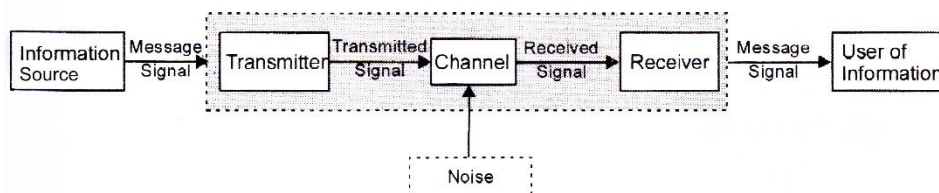
N-S: Concave pole pieces of Horse shoe magnet

C: Soft iron core

(iii) (a) The magnetization of a material may be defined as the net magnetic moment per unit volume of the material.

(b) The magnetic intensity (H) is defined as the product of the number of turns per unit length in a coil (n) and the current that it carries (I). Its S.I. unit is A/m. Hence, $H = nI$.

(iv)



(v) Given:

length, $l = 3.142\text{cm}$

$d = 5.0\text{cm}$

$\Rightarrow r = 2.5 \times 10^{-2}\text{m}$

$i = 5\text{A}$

No. of turns, $N = 500 \times 2$

$= 1000$

Magnetic induction at the centre,

$$\begin{aligned}
 B &= \frac{\mu_0 Ni}{l} \\
 &= \frac{4\pi \times 10^{-7} \times 500 \times 2 \times 5}{3.14} \\
 &= \frac{4\pi \times 3.14 \times 10^{-7} \times 5000}{3.14} \\
 &= 2.0 \times 10^{-3} \text{ T}
 \end{aligned}$$

(vi) Given:

No of turns, $N = 300$

Area, $A = 5 \times 10^{-3} \text{ m}^2$

Current, $I = 15 \text{ A}$

The magnetic moment of the coil,

$$M = NIA$$

$$= 300 \times 15 \times 5 \times 10^{-3}$$

$$M = 22.5 \text{ Am}^2.$$

(vii) Given:

$$\phi = 8t^2 + 6t + C$$

Where, C is a constant, t is in sec, ϕ is in milli-weber

The induced e.m.f. in the loop,

$$|E| = \frac{d\phi}{dt} = \frac{d(8t^2 + 6t + C)}{dt}$$

$$= 16t + 6$$

At $t = 2 \text{ s}$,

$$|E| = \frac{d\phi}{dt}$$

$$= 16 \times (2) + 6$$

$$= 32 + 6$$

$$38 \text{ mV}$$

(viii) We know that,

$$E_n = \frac{E_o}{n^2}$$

Thus,

$$E_5 = \frac{13.6}{5^2}$$

$$\Rightarrow E_5 = 0.544\text{eV}$$

Q 7. Answer any THREE of the following:

[9]

- (i) Obtain an expression for the radius of Bohr orbit for H-atom.
- (ii) What are α and β parameters for a transistor? Obtain a relation between them.
- (iii) Two metal spheres having charge densities $5\mu\text{C}/\text{m}^2$ and $-2\mu\text{C}/\text{m}^2$ with radii 2mm and 1mm respectively are kept in a hypothetical closed surface. Calculate total normal electric induction over the closed surface.
- (iv) The threshold wavelength of silver is $3800\text{\AA}$. Calculate the maximum kinetic energy in eV of photoelectrons emitted, when ultraviolet light of wavelength $2600\text{\AA}$ falls on it. (Planck's constant, $h = 6.63 \times 10^{-34}\text{Js}$, Velocity of light in air, $c = 3 \times 10^8\text{m/s}$)

Solution 7:

(i) Consider an electron revolving around the nucleus in circular orbit of radius r . Let m and e be the mass and charge of the electron and v be the linear speed of the electron. According to Bohr's first postulate,

Centripetal force = electrostatic force of attraction

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2}$$

$$\Rightarrow v^2 = \frac{e^2}{4\pi\epsilon_0 mr} \text{----- (i)}$$

According to Bohr's 2nd postulate,
Angular momentum

$$= n \frac{h}{2\pi}$$

$$\Rightarrow mvr = \frac{nh}{2\pi}$$

$$\Rightarrow v = \frac{nh}{2\pi mr} \text{----- (ii)}$$

$$\Rightarrow v^2 = \frac{n^2 h^2}{4\pi^2 m^2 r^2} \text{----- (iii)}$$

Comparing equation (i) and equation (iii), we get,

$$\frac{e^2}{4\pi\epsilon_0 mr} = \frac{n^2 h^2}{4\pi^2 m^2 r^2}$$

$$\Rightarrow r = \left(\frac{h^2 \epsilon_0}{\pi m e^2} \right) n^2$$

This equation gives the radius of nth Bohr orbit for H-atom.

(ii) The current gain in the common- emitter circuit is called beta(β), Beta is the relationship of collector current to base current. To calculate beta, use the following formula:

$$\beta = \frac{I_C}{I_B}$$

The current gain in the common- base circuit is calculated in a method similar to that of the common emitter except that the input current is I_E not I_B and the term alpha (α) is used in place of beta for gain. Alpha is the ratio of collector current to emitter current. Alpha is calculated by using the formula:

$$\alpha = \frac{I_C}{I_E}$$

Relation between α and β :

We have, $I_E = I_B + I_C$

Now, dividing both sides by I_C , we have,

$$\frac{I_E}{I_C} = \frac{I_B + I_C}{I_C}$$

$$\text{But } \alpha = \frac{I_C}{I_E} \text{ and } \beta = \frac{I_C}{I_B}$$

Thus,

$$\frac{1}{\alpha} = \frac{1}{\beta} + 1$$

$$\frac{1}{\alpha} = \frac{1 + \beta}{\beta}$$

$$\alpha = \frac{\beta}{\beta + 1}$$

(iii) Given,

$$\sigma_1 = 5\mu\text{C} / \text{m}^2$$

$$\sigma_2 = -2\mu\text{C} / \text{m}^2$$

$$r_1 = 2\text{mm}, r_2 = 1\text{mm}$$

Charge enclosed by each sphere,

$$q_1 = 4\pi r_1^2 \times \sigma_1$$

$$q_1 = 4 \times 3.14 \times (2 \times 10^{-3})^2 \times 5 \times 10^{-6}$$

$$q_1 = 4 \times 3.14 \times 20 \times 10^{-12}$$

$$q_1 = 251.3 \times 10^{-12} \text{ C}$$

$$q_2 = 4\pi r_2^2 \times \sigma_2$$

$$q_2 = 4 \times 3.14 \times (10^{-3})^2 \times (-2 \times 10^{-3})^2$$

$$q_2 = 25.13 \times 10^{-12} \text{ C}$$

According to the T.N.E.I. (Total Normal Electric Induction), "it is equal to the total charge enclosed by that surface."

$$\text{T.N.E.I.} = \sum q$$

$$= q_1 + q_2$$

$$= 251.3 \times 10^{-12} + (-25.13 \times 10^{-12})$$

$$= 226.17 \times 10^{-12} \text{ Nm}^2 / \text{C}$$

(iv) It is given that

$$\lambda_o = 3800, \text{ \AA} = 3.80 \times 10^{-7} \text{ m}$$

$$\lambda = 2600, \text{ \AA} = 2.60 \times 10^{-7} \text{ m}$$

We know that,

$$\text{K.E.}_{\text{max}} = hc \left[\frac{1}{\lambda} - \frac{1}{\lambda_o} \right]$$

$$\text{K.E.}_{\text{max}} = 6.63 \times 10^{-34} \times 3 \times 10^8 \times \left[\frac{1}{2.60 \times 10^{-7}} - \frac{1}{3.80 \times 10^{-7}} \right]$$

$$= 1.989 \times 10^{-18} \left[\frac{3.8 - 2.6}{3.8 \times 2.6} \right]$$

$$= 2.4157 \times 10^{-19} \text{ J}$$

$$= \frac{2.4157 \times 10^{-19}}{1.6 \times 10^{-19}} \text{ eV}$$

$$= 1.51 \text{ eV}$$

Q8.

[7]

Obtain an expression for e.m.f. induced in a coil rotating with uniform angular velocity and a uniform magnetic field. Show graphically the variation of e.m.f. with time(t).

Resistance of a potentiometer wire is $0.1\Omega/\text{cm}$. A cell of e.m.f. 1.5V is balanced on 300cm on this potentiometer wire. Calculate the current and balancing length for another cell of e.m.f. 1.4 V on the same potentiometer wire.

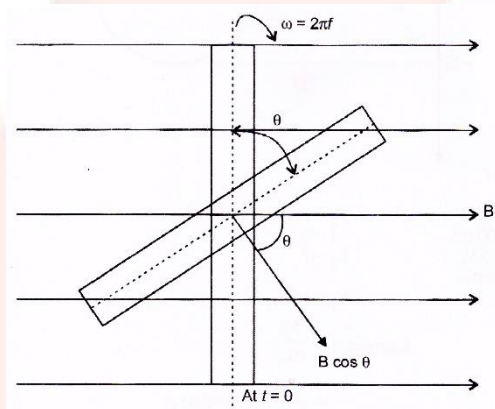
OR

Describe biprism experiment to calculate the wavelength of a monochromatic light. Draw the necessary ray diagram.

If the critical angle of a medium is $\sin^{-1}\left(\frac{3}{5}\right)$, find the polarizing angle.

Solution 8:

Consider a coil of N turns with effective area NA placed in uniform magnetic field as shown in the figure below:



The coil is rotated continuously with constant angular velocity ω . At $t = 0$, initially, the plane of the coil is perpendicular to the magnetic induction B. The magnetic flux passing through the coil is NAB .

Magnetic flux ϕ through the coil at time t is given by,

$$\phi = NBA \cos \theta \text{ -----(i)}$$

The plane of the coil is perpendicular to the magnetic field then after time t, is rotated by angle θ .

Hence,

$$\omega = \frac{\theta}{t}$$

$$\Rightarrow \theta = \omega t$$

By equation (i),

$$\phi = NBA \cos \omega t \text{ -----(ii)}$$

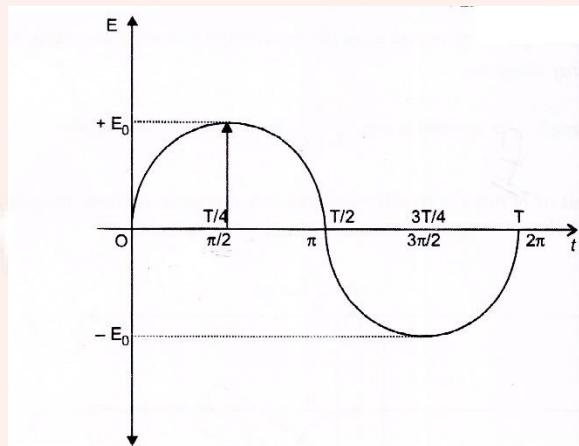
By Faraday's law,

$$E = -\frac{d\phi}{dt} = -\frac{d}{dt}(NBA \cos \omega t)$$

$$\Rightarrow E = NBA\omega \sin \omega t$$

This is an expression for induced e.m.f.

The graph showing the variation with time t of the e.m.f. induced in the coil is given below:



Numerical:

Given:

$$R / L = \sigma = 0.1 \Omega / \text{cm},$$

$$L_1 = 300 \text{cm},$$

$$V_1 = 1.5 \text{V},$$

$$V_2 = 1.4 \text{V}$$

$$I = ?$$

$$I_2 = ?$$

We know that,

$$V_1 = I\sigma L_1$$

$$\therefore \text{Current, } I = \frac{V_1}{\sigma L_1}$$

$$= \frac{1.5}{0.1 \times 300} = 0.05$$

Now,

$$K = \frac{V_1}{L_1} = \frac{1.5}{300} = 5 \times 10^{-3} \text{V / cm}$$

Balancing length,

$$L_2 = \frac{V_2}{K} = \frac{1.4}{5 \times 10^{-3}} = 280 \text{ cm}$$

OR

Distance between the source and eyepiece is directly measured on the scale marked on optical bench. And the micrometer screw fitted to eyepiece is adjusted. And to measure the fringe width β reading of corresponding micrometer is noted. The vertical wire is coincide with 5th, 10th and 15th bright fringes.

Distance d between coherent vertical sources-Conjugate foci method:

Convex lens having the focal length less than $d/4$ is mounted on the stand between eyepiece and bi prism.

For position L_1 of lens:

The distance d_1 between two magnified images is measured by micrometer.

For lens L_2 :

The distance d_2 diminished is measured by moving the lens towards the eyepiece.

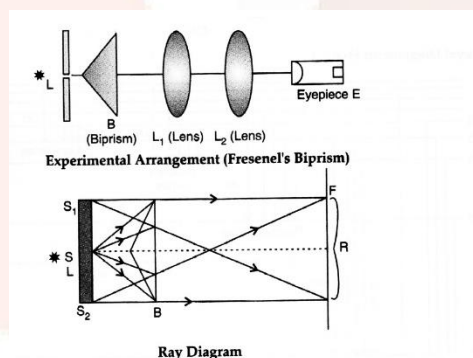
Then,

$$d = \sqrt{d_1 d_2}$$

Therefore, the wavelength is

$$\lambda = \frac{\beta d}{D}$$

$$= \frac{\beta \sqrt{d_1 d_2}}{D}$$



Numerical:

To find the polarizing angle:

$$i_p = \tan^{-1}(\mu)$$

$$C = \sin^{-1}\left(\frac{3}{5}\right)$$

$$\sin C = \left(\frac{3}{5}\right)$$

$$\mu = \frac{1}{\sin C}$$

$$\mu = \frac{5}{3}$$

$$i_p = \tan^{-1}(\mu) = \tan^{-1}\left(\frac{5}{3}\right)$$

$$i_p = 59.03^\circ$$